

MUNDOSTREAMER

DJI FPV Video Receiver & Ground Station

OFFICIAL USER MANUAL • RASPBERRY PI 4B EDITION

Target Hardware	Raspberry Pi 4 Model B (Broadcom BCM2711 64-bit Quad-Core ARM Cortex-A72)
Kernel & OS	Debian 12 Bookworm 64-bit (Linux 6.12 Direct KMS/DRM Plane Engine)
Goggles Compatibility	DJI Goggles V1, V2, Goggles 2, Goggles Integra, Goggles 3 (AOA & Network)
Primary Display Output	Micro-HDMI 0 (HDMI0) 1080p @ 60.0 FPS Zero-Copy Direct DRM Scanout (<1ms lag)
Wireless Connectivity	Onboard Dual-Band Wi-Fi (Cypress CYW43455 2.4/5GHz 802.11ac) & BLE 5.0
DVR Recording Format	Full HD 1080p H.265 (HEVC) / H.264 MP4 Bitstream Storage to USB 3.0 SSD/HDD
Document Version	v2.4.1 Production Release — September 2026

[KEY HIGHLIGHTS] WELCOME TO MUNDOSTREAMER RASPBERRY PI 4B

NEW GENERATION HIGHLIGHTS (RASPBERRY PI 4B EDITION):

- Zero-Copy Direct DRM Scanout (Locked 60.0 FPS): Video frames bypass X11/compositors for razor-sharp 60 FPS playback with zero dropped frames.
- Built-in Dual-Band Wi-Fi Ready: The spectator portal hotspot broadcasts straight from the onboard PCB antenna without extra USB dongles.
- Multi-Flight Diagnostics Exporter: 1-click bundle export of 15 flight logs and kernel metrics for rapid remote analysis via WhatsApp.
- Offline Patch Update System (.mscf / .zip): Upload hotfixes and software patches directly without internet connectivity on the field.
- Instant Safe HDMI Replay: Review recorded flight footage on the big screen without disconnecting the pilot's live DJI USB connection.
- High-Speed USB 3.0 Storage DVR: Lossless Full HD H.265 pure bitstream recording to USB 3.0 portable SSD or fast flash drive.

Keep this manual as your official reference for field operations and appliance maintenance.

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CHAPTER 1: INTRODUCING MUNDOSTREAMER GROUND STATION

MundoStreamer is a standalone, high-performance FPV video receiver station designed specifically for racing drone pilots, freestyle enthusiasts, aerial cinematographers, and event organizers. It bridges high-speed digital video transmissions from the DJI FPV goggles ecosystem to big-screen HDMI displays, laptops, and spectator mobile devices wirelessly.

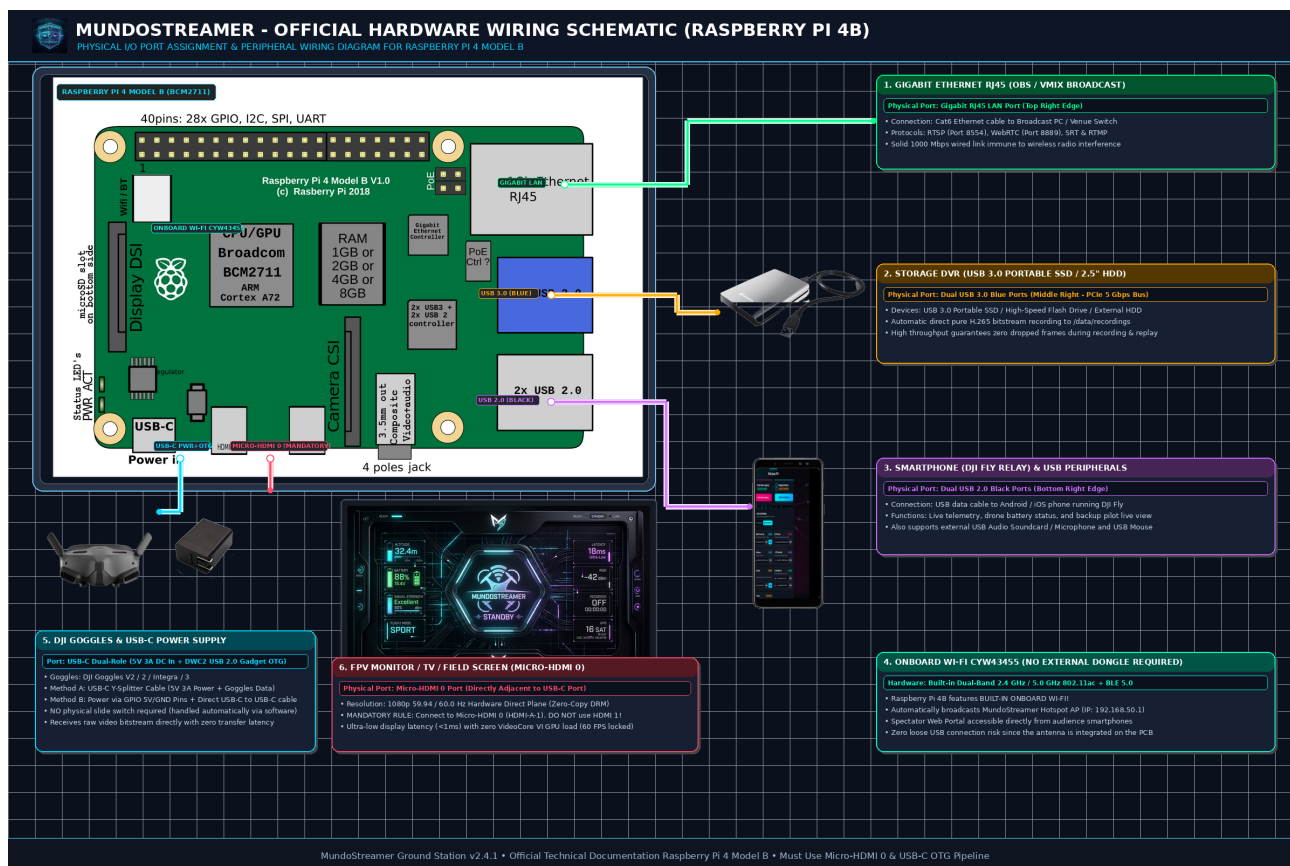
Why MundoStreamer Excels on Raspberry Pi 4B

Unlike conventional mirroring solutions that run on Android or heavyweight PC software stacks with massive latency, MundoStreamer is built on a pure embedded Linux architecture (Broadcom BCM2711 SoC) using hardware-accelerated V4L2-Request decoding and direct Kernel Mode Setting (KMS) video plane scanout (drmModeSetPlane). This architecture delivers decisive advantages:

- **Zero-Copy Direct DRM Scanout:** Raw video packets from the USB-C goggles port decode straight into the hardware display plane without desktop compositing delays (<1 ms display latency).
- **Rock-Solid 1080p @ 60.0 FPS:** The HDMI display is synchronized directly to the 60Hz hardware refresh rate, maintaining a stable 60.0 FPS with zero dropped frames for ultra-fluid drone flight visual tracking.
- **Simultaneous Dual-Stream Broadcasting:** Spectators can enjoy HDMI pit monitors while dozens of mobile spectators stream live via WebRTC/RTSP simultaneously.
- **Pure Bitstream DVR Recording:** Video is captured directly from the camera stream prior to display rendering, producing authentic 1080p H.265 recordings without double compression artifacts.

CHAPTER 2: PHYSICAL PORT LAYOUT & WIRING DIAGRAM

To ensure optimal performance and eliminate cable-induced signal bottlenecks, it is essential to understand the physical I/O port layout on the Raspberry Pi 4 Model B board. The high-resolution technical diagram below illustrates the official setup:



[CONNECTION RULES] PRIMARY CABLING OVERVIEW (RPI 4B)

Field Cabling Quick Summary (Raspberry Pi 4B):

- **USB-C Port (DJI Goggles & Power):** Connect via USB-C Y-Splitter cable (power branch to 5V 3A adapter, data branch to DJI Goggles). USB OTG mode is automatically engaged.
- **Micro-HDMI 0 Port (TV Monitor):** MANDATORY: USE MICRO-HDMI 0 (the port adjacent to the USB-C plug). Connect to field TV or FPV monitor.
- **USB 3.0 Blue Ports (DVR Storage):** Plug your USB 3.0 Portable SSD or fast flash drive here for automatic video recording.
- **USB 2.0 Black Ports (Smartphone / Audio):** Connect to mobile phone for DJI Fly Relay or plug an external USB Soundcard / Microphone.
- **Onboard Wi-Fi:** Broadcasts the MundoStreamer hotspot autonomously without any external USB dongles.

CHAPTER 2 (CONTINUED): PORT SPECIFICATIONS & RULES

The following matrix outlines data bus architectures, kernel drivers, and mandatory usage guidelines for each physical port on Raspberry Pi 4B:

PHYSICAL PORT	IDENTIFICATION	ASSIGNMENT & MANDATORY USAGE
USB Type-C Port (Bottom Left Edge)	DWC2 USB 2.0 OTG & 5V/3A DC	DEDICATED DJI GOGGLES & MAIN POWER IN. Connection: USB-C Y-Splitter cable (5V 3A power + Goggles data). Uses DWC2 hardware peripheral controller for zero-latency data stream. No slide switch required.
Micro-HDMI 0 Port (Adjacent to USB-C)	Direct DRM Plane (HDMI-A-1)	PRIMARY MONITOR OUTPUT (MANDATORY HDMI 0). Connect Micro-HDMI to Full HDMI cable to field TV, pit monitor, or projector. Outputs pure 1080p @ 60.0 FPS (<1ms lag). Do NOT use Micro-HDMI 1!
Micro-HDMI 1 Port (Middle Bottom Edge)	Secondary KMS Output	RESERVED SECONDARY DISPLAY. Reserved for secondary custom displays when configured.
Dual USB 3.0 Blue (Middle Right Stack)	VIA VL805 PCIe Host 5 Gbps	DVR STORAGE & UVC VIDEO CAPTURE. Connect USB 3.0 Portable SSD, fast flash drive, or 2.5" HDD for automatic DVR recordings (path: /data/recordings). Also supports UVC capture cards.
Dual USB 2.0 Black (Bottom Right Stack)	VL805 USB 2.0 Hub Host	DJI FLY PHONE RELAY & USB AUDIO. Connect USB cable to smartphone for DJI Fly (telemetry & status) or plug in external USB Audio Soundcards / commentary microphones.
Gigabit Ethernet RJ45 (Top Right Stack)	BCM54213PE Native Gigabit	OBS / VMIX BROADCAST LAN LINK. Direct Cat6 cable to broadcast PC, vMix switcher, or venue router for interference-free RTSP/WebRTC/SRT feeds.
Onboard Dual-Band Wi-Fi (Internal PCB Module)	Cypress CYW43455 802.11ac	BUILT-IN HOTSPOT AP & WI-FI CLIENT. Raspberry Pi 4B features built-in Wi-Fi! Automatically broadcasts MundoStreamer Hotspot (IP: 192.168.50.1) without external USB dongles.
3.5mm TRRS Audio Jack (Bottom Right Edge)	BCM2711 PWM Stereo DAC	ANALOG STEREO AUDIO LINE OUT. Streams drone pilot telemetry audio to active pit speakers or judge monitoring headphones.

[CRITICAL WARNING] MANDATORY MICRO-HDMI 0 PORT

CRITICAL REQUIREMENT: ALWAYS CONNECT TO MICRO-HDMI 0

Raspberry Pi 4B features two side-by-side Micro-HDMI ports.

- Micro-HDMI 0 (HDMI0) is the port located closest to the USB-C power plug.
- MundoStreamer's zero-copy display pipeline binds strictly to hardware controller HDMI-A-1 (Micro-HDMI 0).
- Connecting to Micro-HDMI 1 will result in a black screen with no video output!

CHAPTER 3: QUICK START GUIDE

Follow these 7 straightforward steps to set up and operate MundoStreamer smoothly on the flight field:

Step 1	Insert MicroSD Golden Master Image into Raspberry Pi 4B Insert a high-speed MicroSD card (Class 10 A1/A2) flashed with the official MundoStreamer image into the underside slot.
Step 2	Connect Video Display to Micro-HDMI 0 Port Plug a Micro-HDMI cable into Micro-HDMI 0 (next to USB-C) and connect the full-size HDMI end to your TV, monitor, or field display.
Step 3	Connect USB-C Y-Splitter & DJI Goggles Plug the male USB-C Y-splitter into the Pi 4B USB-C port. Connect the power branch to a 5V 3A adapter and the USB data branch to your DJI Goggles.
Step 4	Attach DVR Storage to Blue USB 3.0 Port Plug your USB 3.0 Portable SSD or high-speed flash drive into one of the blue USB 3.0 ports (middle stack) for automatic video recording.
Step 5	Power On & Await Standby Animation (~20 Seconds) Supply 5V 3A power. In ~20-25 seconds, the HDMI screen will begin playing the official MundoStreamer standby animation video.
Step 6	Connect Smartphone to MundoStreamer Wi-Fi Hotspot On your phone or laptop, join Wi-Fi network MundoStreamer (Default Password: 1234512345). Open a browser to http://192.168.50.1:8000 or http://mundostreamer.local:8000 .
Step 7	Power Up Drone & Enjoy Pure 60 FPS Live Feed! Power up your drone. The instant camera transmission starts, the HDMI display seamlessly switches to the 1080p @ 60 FPS pure live feed!

HDMI Display Visual Status Indicators

Your HDMI monitor provides instant feedback regarding link status:

- Moving Standby Video Animation: MundoStreamer is healthy and awaiting drone/goggles signal.
- Full-Screen Live Drone Video: Goggles linked and drone active. Zero-latency (<1ms) live feed.
- Black Screen / No Signal: Verify cable is in Micro-HDMI 0 and check that the 5V 3A power supply is adequate.

[FIELD OPERATIONS] INSTANT USB RE-SYNC

INSTANT GOGGLES RE-SYNC TIP:

If you swap goggles batteries or disconnect the USB cable while the unit is running, you do NOT need to reboot! Simply open the Web UI on your phone and tap the blue [Re-sync USB] button on Tab 1 to restore live video instantly.

CHAPTER 4: SMARTPHONE CONNECTION & WEB PORTAL

MundoStreamer offers two modern web interfaces: a lightweight Spectator Landing Page for audience viewing and a comprehensive Admin Control Dashboard for pilot telemetry and technical configuration.

Spectator Landing Page

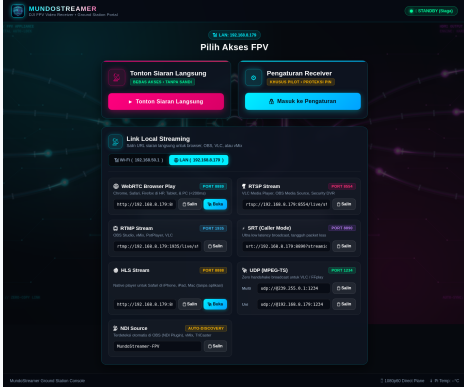


Figure 4.1: Spectator Portal & Low-Latency WebRTC

The default landing page when spectators connect to Wi-Fi. Allows instant viewing of live FPV racing (<200ms latency) without installing any client apps.

Admin Login Interface

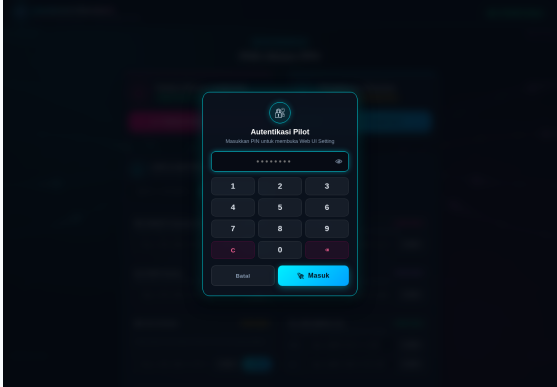


Figure 4.2: Full Access Management Login

Click the Admin Login button in the top-right corner of the portal to enter the full configuration dashboard with your administrative credentials.

Default Credentials & Rapid Access Matrix

Access Parameter	Default Value	Operational Notes
Wi-Fi Hotspot SSID	MundoStreamer	Broadcast directly by RPi 4B onboard Wi-Fi chip
Wi-Fi Password	1234512345	Secure WPA2-PSK, customizable in Tab 9
Web URL (IP Direct)	http://192.168.50.1:8000	Static gateway address of the MundoStreamer hotspot
Web URL (mDNS Domain)	http://mundostreamer.local:8000	Zero-config domain resolution without memorizing IPs
Admin Username	admin	Full administrative management account
Admin Password	11118888	Customizable under Tab 10

CHAPTER 5: WEB UI CONFIGURATION GUIDE

Upon logging into the Admin Dashboard, you have access to 10 modular configuration tabs. Here is the operational breakdown for Tab 1 (Receiver Mode) and Tab 2 (HDMI Output):

Tab 1: Receiver Mode

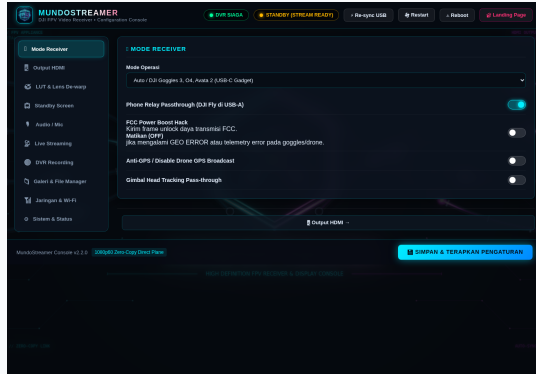


Figure 5.1: DJI Goggles Operating Mode

Features & Options:

- DJI Goggles (Auto): Standard mode for V1, V2, Goggles 2, Integra, and G3.
- UVC Capture Card: Handles external USB video capture inputs.
- Re-sync USB: Instantly re-initializes the USB link without rebooting.
- Reboot System: Performs a clean reboot of the entire appliance.

Tab 2: HDMI Output

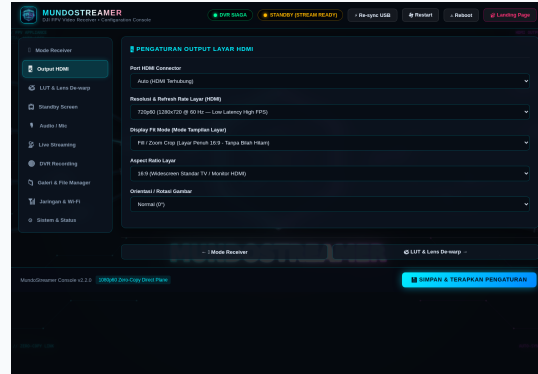


Figure 5.2: HDMI Monitor & Display Tuning

Features & Options:

- Display Resolution: 1080p @ 60 FPS Direct DRM Hardware Plane.
- Fit Mode: Fit (original aspect), Cover (fill screen without black bars), Stretch.
- Aspect Ratio: Preserve 16:9 or 4:3 native camera ratios.
- HDMI Audio: Direct telemetry audio stream to TV speakers.

Emergency Quick Actions in Tab 1

Tab 1 contains emergency recovery buttons crucial for live racing events:

- [Re-sync USB]: Cycles the DWC2 USB gadget OTG link without rebooting the OS. Clears frozen video instantly if a cable was bumped.
- [Restart Streamer]: Re-spawns background receiver engines cleanly in under 2 seconds.
- [Reboot]: Gracefully restarts the Raspberry Pi 4B appliance.

CHAPTER 5: WEB UI CONFIGURATION GUIDE (CONT.)

Configuration overview for Tab 3 (LUT & Lens De-warp) and Tab 4 (Standby Screen Customization):

Tab 3: LUT & De-warp (Hardware Locked)

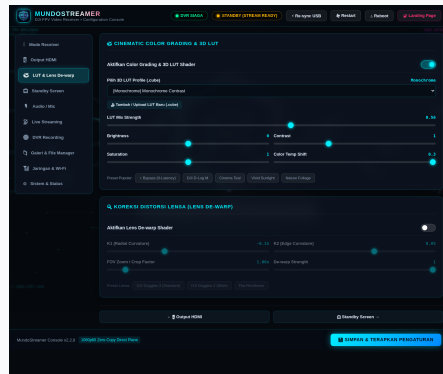


Figure 5.3: Feature Exclusive to Radxa Hardware
Raspberry Pi 4B Specific Status:

- **Hardware Locked:** Tab 3 is permanently locked with banner: '3D LUT & Lens De-warp Exclusive to Radxa Hardware'.
- **GPU Architecture Reason:** The VideoCore VI GPU on BCM2711 is not designed for real-time GLES 3D fragment shaders at 1080p60 (drops to 26–28 FPS).
- **Guaranteed 60.0 FPS:** MundoStreamer routes video through Zero-Copy Direct DRM Scanout for locked 60.0 FPS zero-latency display!

Tab 4: Standby Screen

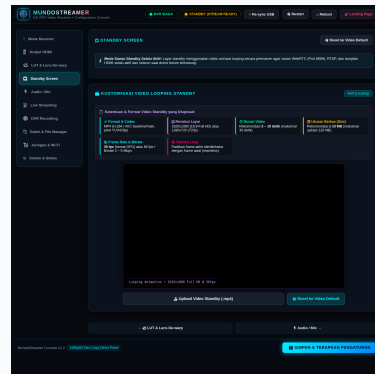


Figure 5.4: Custom Standby Video & Branding

Features & Options:

- **Default Standby Animation:** High-tech official MundoStreamer video loop when idle.
- **Custom Video Upload:** Upload sponsor videos, team logos, or event teasers.
- **Clean Standby Broadcast:** Pure presentation free from watermarks or QR code overlays.

[TECHNICAL DESIGN] ZERO-COPY DIRECT DRM SCANOUT (LOCKED 60 FPS)

ARCHITECTURAL RATIONALE: ZERO-COPY DIRECT DRM SCANOUT

Extensive benchmarking demonstrates that executing real-time fragment shaders on the VideoCore VI GPU introduces severe dropped frames and micro-stuttering.

- MundoStreamer prioritizes maximum framerate stability, zero latency, and pure 60.0 FPS via direct kernel DRM plane scanning.
- If your production demands real-time fisheye lens de-warping and 3D LUT color grading on big-screen displays, choose the Radxa ROCK 3A edition powered by the ARM Mali-G52 GPU.

CHAPTER 5: WEB UI CONFIGURATION GUIDE (CONT.)

Configuration breakdown for Tab 5 (Audio Management) and Tab 6 (Live Streaming Broadcasts):

Tab 5: Audio Configuration

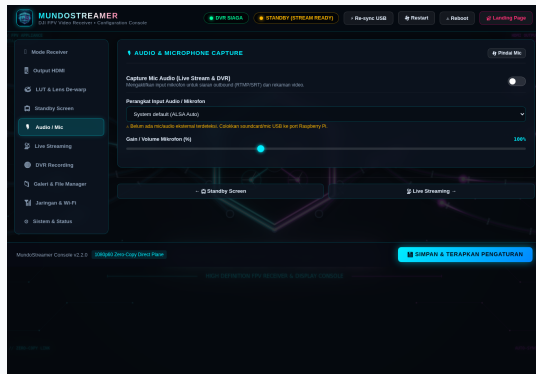


Figure 5.5: Sound Output & Mic Inputs

Features & Options:

- HDMI Audio Out: Routes pilot telemetry audio to TV speakers via Micro-HDMI 0.
- 3.5mm Stereo Out: Analog output for headphones and judges' monitor speakers.
- USB Microphone Input: Plug USB soundcard/mic into USB 2.0 for live commentary.
- Master Volume & Boost: Clean audio gain controls.

Tab 6: Live Streaming

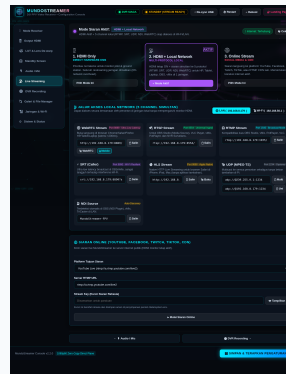


Figure 5.6: Broadcast Protocols & OBS Link

Features & Options:

- RTSP Server (Port 8554): HD video feed for OBS Studio and vMix mixers.
- WebRTC Spectator (Port 8889): Real-time (<200ms) spectator viewing in phone browsers.
- RTMP Cloud Push: Direct streaming to YouTube, TikTok, or Facebook Live.
- SRT Protocol: Encrypted remote transmission with packet-loss recovery.

Simultaneous Multi-Channel Broadcast Pipeline

MundoStreamer enables concurrent distribution pathways without bogging down the Raspberry Pi 4B processor:

1. Main Stage HDMI (Micro-HDMI 0): Zero-latency display for race directors, commentators, and audience monitors.
2. Wireless Spectator Viewing (WebRTC): Pit area spectators watch directly in their phone browsers with <200ms lag.
3. Broadcast Production Feed (Gigabit LAN): High-speed Cat6 Ethernet to your OBS/vMix laptop for graphics mixing and cloud broadcasting.

CHAPTER 5: WEB UI CONFIGURATION GUIDE (CONT.)

Configuration breakdown for Tab 7 (DVR Recording) and Tab 8 (Video Gallery & Instant HDMI Replay):

Tab 7: DVR Recording

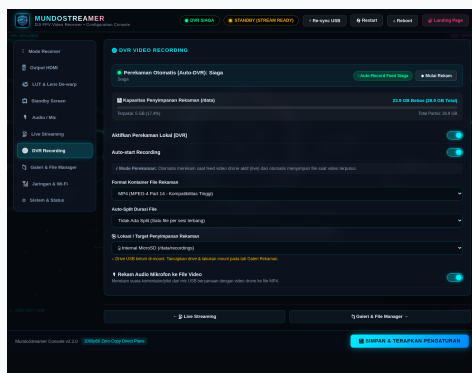


Figure 5.7: Automatic DVR Recording Setup

Features & Options:

- Auto-Start Recording: Automatically triggers recording when the drone arms.
- Pure Bitstream Capture: Stores native Full HD H.265/H.264 without re-encoding.
- File Segmentation: Segment video every 5 or 10 minutes to protect against accidental power loss.

Tab 8: Video Gallery & Replay

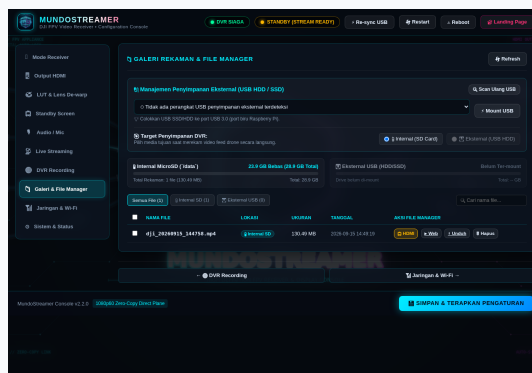


Figure 5.8: Recording Gallery & HDMI Replay

Features & Options:

- **Fast Loading (48 ms):** Instant access to dozens of videos via native MP4 atom parser.
- **Play on HDMI Button:** Replay flights on the pit TV without disconnecting USB.
- **Flight Safety Interlock:** Instantly blocks/halts replay when pilot flies live!
- **Download & Delete:** Save recordings directly to your phone.

[FEATURE SPOTLIGHT] INSTANT HDMI REPLAY & FLIGHT SAFETY INTERLOCK

INSTANT HDMI REPLAY & FLIGHT SAFETY INTERLOCK:

On Tab 8, click the blue [Play on HDMI] button next to any flight recording to review it on the big screen.

- **Flight Safety Interlock:** If a pilot unexpectedly arms and takes off while a replay is running, the safety sensor instantly terminates playback within milliseconds and switches the screen back to the live flight feed!

CHAPTER 5: WEB UI CONFIGURATION GUIDE (CONT.)

Configuration breakdown for Tab 9 (Network & Wi-Fi) and Tab 10 (System, Diagnostics & Offline Patch):

Tab 9: Network & Wi-Fi

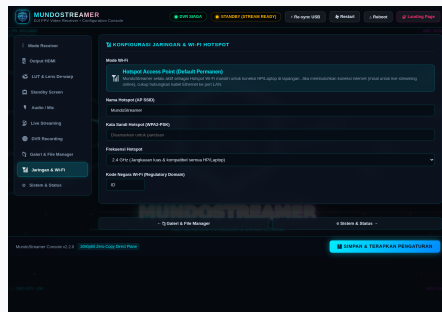


Figure 5.9: Wi-Fi Hotspot & Client Settings

Features & Options:

- Onboard Hotspot AP: Configure SSID, password, and 2.4/5GHz channels.
- Wi-Fi Client Mode: Connect to venue Wi-Fi or phone hotspot for internet access.
- IP Status Display: Shows IP addresses for both Wi-Fi and wired LAN.

Tab 10: System & Diagnostics

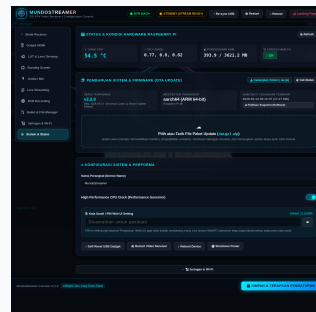


Figure 5.10: Diagnostics, Licensing & Power

New Features in v2.4.1:

- Export Diagnostic Logs: 1-click bundle export of 15 flight logs for WhatsApp support.
- Offline Patch Update (.mscf / .zip): Upload hotfix packages with no internet needed.
- Hardware Telemetry: Real-time BCM2711 CPU temperature and RAM stats.
- Power Off System: Clean shutdown prior to power disconnection.

[BUILT-IN HARDWARE] READY-TO-USE ONBOARD WI-FI

ONBOARD WI-FI ADVANTAGE ON RASPBERRY PI 4B:

Raspberry Pi 4 Model B is equipped with a built-in Cypress CYW43455 wireless controller supporting Dual-Band 2.4 GHz and 5.0 GHz 802.11ac.

- You do not need an external USB Wi-Fi dongle. The MundoStreamer hotspot is active immediately upon boot.
- For extreme outdoor distances (>100 meters), you may still attach a high-power external USB Wi-Fi adapter to the USB 2.0 port.

[OPERATIONAL SAFETY] POWER RESILIENCY & SHUTDOWN SOP

POWER RESILIENCY & PROPER SHUTDOWN PROCEDURE:

MundoStreamer operates with a Persistent Read-Only Rootfs (/ro) overlay architecture, rendering the system 100% immune to sudden power cuts without filesystem corruption risk.

- Field Shutdown SOP: Navigate to Tab 10: System and click the red Power Off System button. Wait 10-15 seconds until the HDMI screen goes black and LEDs stop blinking before disconnecting power.

CHAPTER 6: DVR RECORDING & HDMI REPLAY WORKFLOW

MundoStreamer features an industrial-grade DVR subsystem engineered to guarantee that competitive racing and cinematic flights are saved with pristine quality without impacting live goggles transmission.

1. Zero-Drop Pure Bitstream DVR Pipeline

The DVR engine leverages a native Tee Splitter Pipeline (`mando_stream_tee`). Raw H.265/H.264 video packets stream directly to USB 3.0 storage the moment they arrive from the USB-C OTG controller. Because no software re-encoding occurs:

- Raspberry Pi 4B CPU load remains minimal (<10%).
- Recordings are 100% identical in quality to the drone's airborne camera (up to 50 Mbps bitstream).
- Zero frame drops occur even during high-velocity maneuvers.

2. Recommended USB 3.0 Storage Media

For peak recording performance and smooth instant playback:

- Use a USB 3.0 Portable SSD or High-Speed USB 3.0 Flash Drive (write speed >60 MB/s).
- Always connect to the Blue USB 3.0 Ports (Middle Stack) to benefit from 5 Gbps PCIe bandwidth.
- Recommended filesystem formats are exFAT (for universal Windows/macOS compatibility) or ext4.

3. Instant Playback on HDMI & Safety Sensor

MundoStreamer allows instant video replay on the big-screen HDMI monitor without disconnecting the pilot's goggles:

Stage	Operational Replay Steps
Step 1	Open the Web UI on your smartphone or laptop and navigate to Tab 8: Video Gallery.
Step 2	Locate the flight recording you wish to review with judges or spectators.
Step 3	Click the blue [Play on HDMI] button next to the recording filename.
Step 4	The video will immediately play on the HDMI display at smooth 60 FPS using BCM2711 hardware acceleration.
Step 5	To conclude playback and return to the standby animation, click [Stop HDMI Playback].

[FLIGHT SAFETY] SAFETY SENSOR INTERLOCK

FLIGHT SAFETY SENSOR INTERLOCK:

MundoStreamer incorporates an automated flight safety sensor: The instant live transmission from a drone is detected, any active HDMI video replay is immediately halted and the display switches to the live FPV feed! This guarantees that live flights are never obscured by recorded video.

CHAPTER 7: LIVE STREAMING TO OBS & VMIX

MundoStreamer provides a full broadcast protocol bridge to integrate DJI goggles video feeds into professional video mixers such as OBS Studio, vMix, Wirecast, or hardware broadcast switchers.

1. RTSP Streaming to OBS Studio / vMix (Recommended)

The RTSP protocol delivers outstanding Full HD picture quality with robust audio sync:

1. Ensure your broadcast laptop is on the same MundoStreamer Wi-Fi or wired LAN network.
2. Open OBS Studio -> Click + (Add Source) -> Select Media Source.
3. Uncheck Local File.
4. In the Input field, enter: `rtsp://192.168.50.1:8554/live`
5. Set Network Buffering to 1 MB or 0 MB (Ultra Low Latency).
6. Click OK. The live drone video will appear immediately on your OBS canvas!

2. WebRTC Live View on Mobile Browsers (No App Required)

Spectators and race judges can watch the live race stream on their mobile devices without installing any apps:

- Connect your phone to Wi-Fi network MundoStreamer.
- Open any web browser (Chrome, Safari, Firefox) and go to: `http://192.168.50.1:8000`
- The live FPV video streams instantly in HD with latency under 200 milliseconds!

Error response
Error code: 404
Message: Not Found.
Error code explanation: 404 - Nothing matches the given URL.

Figure 7.1: Mobile Spectator WebRTC Player Interface

3. Direct RTMP Streaming to YouTube, TikTok & Facebook

When MundoStreamer has internet access (via Wi-Fi client on Tab 9 or Gigabit LAN):

- Navigate to Tab 6: Live Streaming.
- Enter the RTMP URL and Stream Key provided by your streaming platform.
- Enable Enable RTMP Stream and click Save Settings.
- MundoStreamer broadcasts directly to cloud streaming platforms autonomously!

CHAPTER 8: AUDIO COMMENTARY & TRRS PINOUT

MundoStreamer on Raspberry Pi 4 Model B supports multi-path audio routing: HDMI embedded audio, 3.5mm analog headphone monitoring, and external commentator microphone input via USB Audio.

1. Raspberry Pi 4B 3.5mm TRRS Port Pinout Standard

The onboard 3.5mm TRRS jack on the Raspberry Pi 4B provides analog stereo audio output:

Jack Pole	Audio / Video Signal	Functional Description
Tip	Audio Left Out	Left channel stereo output for headphones / monitor speakers
Ring 1	Audio Right Out	Right channel stereo output for headphones / monitor speakers
Ring 2	Ground (GND)	Common audio ground reference
Sleeve	Composite Video Out	Analog composite video signal (CVBS)

2. Adding Commentator Microphones via USB Audio

The onboard 3.5mm jack is dedicated to audio output. To integrate commentator voice into recordings and live streams:

1. Plug a standard USB Microphone (or USB Audio Soundcard adapter) into one of the Black USB 2.0 ports.
2. Open the Web UI -> Navigate to Tab 5: Audio Configuration.
3. Select the USB microphone from the Audio Input Device dropdown.
4. Adjust audio level with the Microphone Boost slider (+15dB to +20dB recommended).
5. Navigate to Tab 7: DVR Recording -> Ensure Audio Mode is set to Video + Mic Audio.
6. Commentary voice will blend seamlessly into Full HD MP4 recordings and RTSP/WebRTC streams!

[AUDIO TIPS] FIELD COMMENTARY MICROPHONE SETUP

BROADCAST AUDIO RECOMMENDATIONS:

- Use a wireless 2.4GHz USB lavalier microphone (such as Hollyland Lark or DJI Mic with USB-A receiver) for freedom of movement around the race track.
- Drone audio from the DJI Goggles will route directly to TV speakers via Micro-HDMI 0 without additional cabling.

CHAPTER 9: MAINTENANCE & TROUBLESHOOTING

Below is a diagnostic resolution table for field situations along with the official procedure for exporting multi-flight diagnostic log bundles and applying offline hotfixes:

ISSUE SYMPTOM	PROBABLE CAUSES	IMMEDIATE FIELD REMEDY
HDMI Screen Dark / No Signal	- Cable plugged into Micro-HDMI 1. - Loose HDMI cable. - Power supply below 5V 3A.	- Move cable to Micro-HDMI 0 (next to USB-C). - Replace with high-quality Micro-HDMI cable. - Verify adapter provides at least 5.0V / 3.0A pure DC.
Display Stays on Standby Video	- USB cable is charge-only (no data). - Drone not yet armed/powered. - USB link not synchronized.	- Ensure USB cable supports high-speed data. - Open Web UI Tab 1 and click [Re-sync USB]. - Ensure DJI Goggles are bound and drone is armed.
BCM2711 CPU Warm to the Touch	- Operating under direct sunlight. - Fan/heatsink dirty or blocked.	- Use an active cooling aluminum case with a 5V fan. - Normal operating temperature is 45°C - 65°C. - Keep appliance shaded during outdoor summer events.
Wi-Fi Hotspot Not Visible on Phone	- Wireless channel incompatible. - Radio set to Client mode.	- Open Wi-Fi settings in Tab 9. - Ensure Mode is set to Access Point (AP). - Set Country Code to US or ID on standard 2.4GHz channels.

[TECHNICAL ASSISTANCE] DIAGNOSTICS & OFFLINE PATCHING SOP

OFFICIAL TECHNICAL SUPPORT & MULTI-FLIGHT DIAGNOSTICS EXPORT:

If an unexpected issue occurs, follow this official diagnostic resolution workflow:

- Export Diagnostic Logs: Open Tab 10: System -> Click the green [Export Diagnostics Logs] button. A complete ZIP archive of up to 15 flight logs and metrics will download to your device.
- Send via WhatsApp Support: Forward the ZIP file to the MundoStreamer Technical Team at: +62 856-4722-9919.
- Receive Offline Patch (.mscf / .zip): Our engineers will analyze the cause and provide an offline hotfix package.
- Apply Update: In Tab 10, select the hotfix file -> Click [Apply Update]. System restores cleanly in 5 seconds with zero internet needed!

CHAPTER 10: LICENSING & 5-LAYER PROTECTION

MundoStreamer is safeguarded by an enterprise-grade ****5-Layer Cryptographic Security Architecture**** engineered to guarantee binary authenticity, firmware integrity, and maximum operating stability in the field.

MundoStreamer 5-Layer Security Architecture

- Layer 1 (Ed25519 Root-of-Trust): Asymmetric cryptographic keys bound to the BCM2711 CPU hardware serial and RPi4B board tag. Card-Swap Freedom: Freely replace or upgrade MicroSD cards without license loss.
- Layer 2 (Cython Native Stripped Binary): Server backend compiled to stripped native ELF 64-bit machine binaries without plain Python runtime scripts.
- Layer 3 (Sentinel Gatekeeper & Barcode Lockscreen): Monitored by the mundo-guard daemon. Displays a Full HD activation QR screen on HDMI if unlicensed.
- Layer 4 (Anti-Tamper & Binary Integrity Verification): Self-verifying SHA-256 hash checks prevent unauthorized alterations from destabilizing video pipelines.
- Layer 5 (Zero Upstream Leakage & Fully Disguised Processes): Streamer processes are fully disguised (mundo_receiver, mundo_stream_tee, mundo_kms) free of third-party remnants.

New Board License Activation Procedure

When inserting the MundoStreamer Golden Master Image into a new Raspberry Pi 4B board, Sentinel Guard will display an activation QR barcode on the HDMI display. Simply reach out via WhatsApp with a photo of your Machine ID to receive your instant activation token.

Official Support Channels	Contact & Communication Details
WhatsApp Technical Support	+62 856-4722-9919 (Fast Response & Licensing)
Telegram FPV Community	@MundoStreamerSupport
Firmware & Patch Portal	http://mundostreamer.local:8000 (Tab 10 System)
Support Operating Hours	Monday – Sunday: 08:00 – 22:00 WIB (Event Ready)

[OUR DEDICATION] FLY SAFE, BROADCAST FAST

THANK YOU FOR CHOOSING MUNDOSTREAMER:

MundoStreamer is dedicated to advancing the sport of FPV drone racing and aerial cinematography worldwide. Experience true zero-latency live feeds, pure Full HD bitstream recordings, and effortless event broadcasting on every flight. Fly Safe, Broadcast Fast!